

BRAC's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: C Batch: 2 Practical No: 1

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Subject Name : Deep Learning

Problem Statement : Install and configure TensorFlow/Keras in Google Colab. Perform data preprocessing, normalization, train-test split, and visualization on a sample dataset.

Algorithm :

1. Import TensorFlow/Keras and required libraries.
2. Load the MNIST dataset.
3. Split into training (60000) and testing (10000) data.
4. Normalize images by dividing pixel values by 255.0.
5. Build a Sequential model (Flatten → Dense(128, ReLU) → Dense(10, Softmax)).
6. Compile with Adam optimizer.
7. Train for 5 epochs.
8. Evaluate the model.
9. Display test accuracy.

Code :

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
import matplotlib.pyplot as plt
```

```
(train_images, train_labels), (test_images, test_labels) = tf.keras.datasets.mnist.load_data()
```

```
train_images = train_images / 255.0
test_images = test_images / 255.0
print(train_images.shape)
print(test_images.shape)
```

```

my_model = tf.keras.models.Sequential()
my_model.add(tf.keras.layers.Flatten(input_shape=(28, 28)))
my_model.add(tf.keras.layers.Dense(128, activation='relu'))
my_model.add(tf.keras.layers.Dense(10, activation='softmax'))
my_model.compile(optimizer='adam', loss='sparse_categorical_crossentropy',
metrics=['accuracy'])
my_model.fit(train_images, train_labels, epochs=5)

val_loss, val_acc = my_model.evaluate(test_images, test_labels)
print('Test accuracy: ', val_acc)

```

Output :

```

[7] 0s #import
import tensorflow as tf
from tensorflow.keras.datasets import mnist
import matplotlib.pyplot as plt

[8] #load the dataset
(train_images, train_labels), (test_images, test_labels) = tf.keras.datasets.fashion_mnist.load_data()

Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz
29515/29515 — 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-images-idx3-ubyte.gz
26421880/26421880 — 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels-idx1-ubyte.gz
5148/5148 — 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images-idx3-ubyte.gz
4422102/4422102 — 0s 0us/step

[9] 0s train_images = train_images / 255.0
test_images = test_images / 255.0

[10] 0s print(train_images.shape)
print(test_images.shape)

... (60000, 28, 28)
(10000, 28, 28)

[11] 25s my_model = tf.keras.models.Sequential()
my_model.add(tf.keras.layers.Flatten(input_shape=(28, 28)))
my_model.add(tf.keras.layers.Dense(128, activation='relu'))
my_model.add(tf.keras.layers.Dense(10, activation='softmax'))

my_model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
my_model.fit(train_images, train_labels, epochs=5)

/usr/local/lib/python3.12/dist-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an "input_shape" / "input_dim" argument to a layer. When using Sequential models, prefer us
super().__init__(**kwargs)
Epoch 1/5
1875/1875 — 7s 3ms/step - accuracy: 0.8257 - loss: 0.4923
Epoch 2/5
1875/1875 — 4s 2ms/step - accuracy: 0.8657 - loss: 0.3698
Epoch 3/5
1875/1875 — 4s 2ms/step - accuracy: 0.8781 - loss: 0.3334
Epoch 4/5
1875/1875 — 5s 3ms/step - accuracy: 0.8852 - loss: 0.3100
Epoch 5/5
1875/1875 — 4s 2ms/step - accuracy: 0.8923 - loss: 0.2921
<keras.src.callbacks.history.History at 0x7f16fc14ad50>

[12] 1s val_loss, val_acc = my_model.evaluate(test_images, test_labels)
print('Test accuracy: ', val_acc)

... 313/313 — 2s 4ms/step - accuracy: 0.8720 - loss: 0.3483
Test accuracy: 0.871999979019165

```